

Technology Stack

Date	27 June 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML, CSS, Jinja2 Templates	Used to create a simple, responsive, and user-friendly interface for entering HDI indicators and displaying prediction results.

2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python, Flask	Flask is a lightweight web framework used to handle user requests, integrate the machine learning model, and generate predictions efficiently.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	CSV File (HDI.csv)	The CSV file is used to store and manage the HDI dataset for training and prediction purposes. It provides simple and efficient data handling for machine learning tasks.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Flask Local Server	The application is deployed and tested using the Flask local server within the Anaconda environment, enabling easy development and execution.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub	GitHub is used for source codemanagement, collaboration among team members, and maintaining version history.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	NumPy, Pandas, Scikit-learn,	NumPy and Pandas are used for data manipulation, Scikit-learn for building the Linear Regression model, Matplotlib and Seaborn for visualization, while Anaconda Navigator and PyCharm support development and package management.

